

AFRICA DIGITAL DEVELOPMENT ANALYSIS (2000–2024)

Exploring the Relationship Between Internet Adoption, Mobile Connectivity, and Economic Growth Using World Bank Development Indicators

Objective / Problem Statement

Africa has experienced rapid growth in digital technologies over the past two decades. However, Internet accessibility, mobile connectivity, and economic development differ significantly across countries.

This project aims to analyse digital development trends across 54 African countries between 2000 and 2024 using the World Bank Development Indicators (WDI). The analysis focuses on understanding Internet adoption, mobile subscriptions, GDP per capita, and the relationships between these indicators to support informed policy and investment decisions.

Dataset Description

Source: World Bank – World Development Indicators (WDI)

The dataset contains development statistics for countries worldwide across multiple sectors, including:

- Economy
- Technology
- Population
- Education
- Health
- Environment

For this project, the analysis focused on African countries and the following indicators:

- Individuals using the Internet (% of population)
- Mobile cellular subscriptions (per 100 people)

- GDP per capita (current US\$)
- GDP growth (annual %)

Coverage:

- 54 African Countries
- 2000–2024

Data Cleaning Process

The following data preparation steps were performed using Python (Pandas):

- Imported and explored the dataset.
- Filtered records to include only African countries.
- Selected relevant development indicators.
- Converted the Year column to numeric format.
- Removed duplicate records.
- Replaced placeholder values (0) with missing values where appropriate.
- Removed records with missing indicator values.
- Merged the development dataset with country metadata (Region and Income Group).
- Created analysis-ready datasets for Internet, Mobile, GDP, and GDP Growth.
- Exported the cleaned dataset for Power BI visualization.

Analysis Methodology

The project followed a complete data analytics workflow:

- Data Collection
- Data Cleaning
- Data Transformation
- Exploratory Data Analysis (EDA)

- KPI Development
- Correlation Analysis
- Trend Analysis
- Dashboard Development in Power BI
- Insight Generation

Tools Used:

- Python (Pandas, NumPy, Matplotlib)
- Power BI
- Microsoft Excel
- GitHub

Key Findings

- The study covered all **54 African countries** between 2000 and 2024.
- Average Internet usage reached **44.89%**, showing significant growth over time.
- Mobile subscriptions averaged **107.25 per 100 people**, indicating widespread multiple SIM ownership and strong mobile market penetration
- Average GDP per capita across the selected countries was **\$2.31K**.
- Internet usage and GDP per capita showed a **strong positive correlation (0.72)**.
- Mobile subscriptions and Internet usage exhibited a **moderate positive correlation (0.50)**.
- Countries such as **Morocco, Seychelles, Mauritius, Libya, and South Africa** ranked among the leaders in digital development.

Insights

- Internet adoption has increased steadily across Africa over the last two decades.
- Mobile technology has become the primary means of digital access.
- Higher Internet penetration is generally associated with stronger economic performance.
- Mobile connectivity supports Internet adoption, although additional factors such as affordability and infrastructure influence Internet usage.
- Digital development remains uneven across African countries, indicating a persistent digital divide.

Recommendations

- Expand broadband infrastructure, particularly in underserved areas.
- Improve Internet affordability through competitive pricing and supportive policies.
- Strengthen digital literacy and ICT education.
- Encourage innovation and investment in digital technologies.
- Leverage Africa's high mobile penetration to expand digital financial services, e-commerce, and e-government solutions.
- Use data-driven monitoring to evaluate progress and guide future digital development strategies.

Conclusion

The analysis demonstrates that digital transformation is progressing across Africa, with significant improvements in Internet adoption and mobile connectivity between 2000 and 2024. Countries with greater digital connectivity generally exhibit stronger economic performance, highlighting the importance of investing in digital infrastructure and inclusive technology policies. While progress has been substantial, continued efforts are needed to reduce the digital divide and ensure that all countries benefit from Africa's digital future.